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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
HU103 - DESIGN THINKING AND INNOVATION

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. The 6-3-5 method is related to: (1) C02 BL2 PO3
 (a) Prototyping (b) Brainstorming
 (c) Testing (d) Strategy formulation
2. The function of business process models is mainly to_____. (1) C02 BL2 PO3
 (a) create new business concepts (b) create substitutes for changed ideas
 (c) visually represent steps and flows of a business operation or process (d) none of these
3. NABC stands for: (1) C02 BL1 PO3
 (a) Need, Approach, Benefit, Competition (b) Network, Analysis, Business, Control
 (c) Need, Analysis, Business, Cost (d) None
4. Which phase uses stakeholder mapping? (1) C01 BL2 PO2
 (a) Define (b) Empathize
 (c) Test (d) Prototype
5. Storytelling is mainly used in: (1) C01 BL2 PO4
 (a) Empathize (b) Define
 (c) Test (d) Prototype

6. Collaboration in distributed design works through_____. (1) C02 BL2 PO3
 (a) shared digital environment (b) real-time communication
 (c) open access (d) all of these
7. The use of scenario based prototyping_____. (1) C02 BL2 PO3
 (a) tests how users navigate the system in realistic situations (b) helps stakeholders visualize the experience
 (c) provides contexts and constraints for users interaction with the prototype (d) none of these
8. Extreme competition leads to: (1) C03 BL2 PO7
 (a) Innovation (b) Saturation
 (c) Monopoly (d) Loss of users
9. Strategic foresight is related to: (1) C03 BL2 PO7
 (a) Testing (b) Growth and innovation
 (c) Prototyping (d) Empathy
10. Which is NOT an ideation tool? (1) C02 BL2 PO3
 (a) Brainstorming (b) 6-3-5
 (c) SCAMPER (d) A/B Testing
11. UX mainly focuses on: (1) C02 BL1 PO1
 (a) Visual appeal (b) User experience
 (c) Backend design (d) Cost reduction
12. Real-time knowledge and capture in design involves_____. (1) C02 BL1 PO3
 (a) capturing user actions and system responses (b) analyzing usability issues and engagement patterns
 (c) understading user experience and product performance (d) all of these

Part B**(Answer any four of the following) 5x4=20**

13. Explain the 2x2 matrix technique with an example. (5) C02 BL3 PO3
14. Explain MVP with an example. (5) C02 BL3 PO5
15. Discuss the role of collaboration in digital design tools. (5) C03 BL3 PO8
16. What is the business model canvas? How does it work? (5) C03 BL3 PO4

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| 17. What is TRIZ? How does it work? | (5) | C02 | BL3 | PO3 |
| 18. Explain the role of real-time design interaction tools in collaborative design environments. | (5) | C01 | BL4 | PO3 |

Part C
(Answer any three of the following) 3x10=30

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| 19. Discuss ideation techniques and their role in creative problem solving. | (10) | C02 | BL5 | PO3 |
| 20. Explain Empathize and Define phases with suitable tools and examples. | (10) | C01 | BL4 | PO2 |
| 21. Answer the following questions. | | | | |
| (a) Discuss some of the principal Tools used in Design Thinking. Give relevant examples. | (8) | C02 | BL3 | PO3 |
| (b) How do they reshape processes with a new Perspective? | (2) | C02 | BL3 | PO3 |
| 22. How do the adoption of Agile principles in a virtual collaboration environment foster team cohesion and keep work in sync? Explain in details. | (10) | C03 | BL4 | PO6 |
| 23. Explain the concept of strategic innovation and discuss how Design Thinking acts as a framework to achieve strategic innovation in organizations. Illustrate with suitable examples. | (10) | C01 | BL2 | PO1 |